H **IATIONAL JUNIOR COLLEGE**

ENIOR HIGH 2 PRELIMINARY EXAMINATION

Higher 2

CANDIDATE NAME			
SUBJECT CLASS		REGISTRATION NUMBER	

PHYSICS **9478/02**

21 Aug 2026
2 hour

Candidate answers on the Question Paper.

No Additional Materials are required.

READ THE INSTRUCTION FIRST

Write your subject class, registration number and name in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.
Answers **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 7
2	/ 7
3	/ 8
4	/ 8
5	/ 8
6	/ 8
7	/ 9
8	/ 20
Deduction	
Total	/75

[Turn over

This document contains **21** printed pages and **3** blank pages.

DATA AND FORMULAE

Data

speed of light in free space	$c = 3.00 \times 10^{8}\ \text{m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\ \text{H m}^{-1}$
permittivity of free space	$= 8.85 \times 10^{-12}\ \text{F m}^{-1}$
	$\left(\quad = 8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
elementary charge	$e = 1.60 \times 10^{-19}\ \text{C}$
Planck constant	$h = 6.63 \times 10^{-34}\ \text{J s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\ \text{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\ \text{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\ \text{kg}$
molar gas constant	$R = 8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Avogadro constant	$N_A = 6.02 \times 10^{23}\ \text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
acceleration of free fall	$g = 9.81\ \text{m s}^{-2}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\,\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	$T/\text{K} = T/^{\circ}\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$

[Turn over

displacement of particle in s.h.m. $x = x_0 \sin \omega t$

velocity of particle in s.h.m. $v = v_0 \cos \omega t = \pm \omega \sqrt{(x_0^2 - x^2)}$

electric current $I = nAvq$

resistors in series $R = R_1 + R_2 + \ldots$

resistors in parallel $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

capacitors in series $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$

capacitors in parallel $C = C_1 + C_2 + \ldots$

energy in a capacitor $U = \frac{1}{2} QV = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$

charging a capacitor $Q = Q_0 \left[1 - e^{-\frac{t}{\tau}}\right]$

discharging a capacitor $Q = Q_0 e^{-\frac{t}{\tau}}$

RC time constant $\tau = RC$

electric potential $V = \frac{Q}{4\pi\varepsilon_0 r}$

alternating current / voltage $x = x_0 \sin \omega t$

magnetic flux density due to a long straight wire $B = \frac{\mu_0 I}{2\pi d}$

magnetic flux density due to a flat circular coil $B = \frac{\mu_0 NI}{2r}$

magnetic flux density due to a long solenoid $B = \mu_0 nI$

energy states for quantum particle in a box $E_n = \frac{h^2}{8mL^2} n^2$

radioactive decay $x = x_0 e^{-\lambda t}$

radioactive decay constant $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 **(a)** State what is meant by the *moment of a force*.

……

……

…… [1]

(b) A traffic light hangs from a uniform metal pole AB, freely pivoted at point A, as shown in Fig. 1.1. The pole is 6.5 m long and has a mass of 9.0 kg. The mass of the traffic light is 11.0 kg.

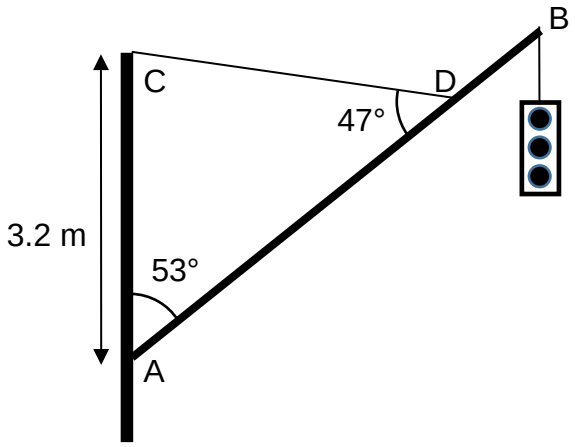


Fig. 1.1

(i) Show that the tension in the cable CD is 250 N.

[2]

(ii) Determine the magnitude of the force exerted on the metal pole at point A.

force = …….. N [2]

(iii) Explain how the magnitude of the force at the point A changes if the traffic light is shifted from B to D.

……..

……..

……..

…… [2]

[Total: 7]

2 A soft rubber ball A of mass 0.50 kg is dropped from a height. At the same time another soft rubber ball B of mass 1.0 kg is projected vertically upward.

The two balls meet and collide head-on.

Fig. 2.1 shows the two balls just before the contact. Ball A is travelling downward at 2.0 m s^{-1} and ball is travelling upward at 1.0 m s^{-1}.

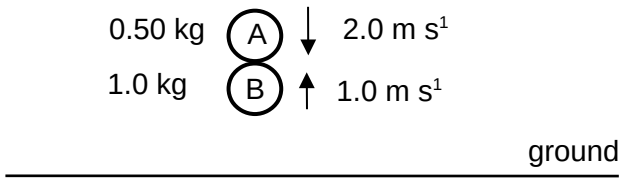


Fig. 2.1 (not to scale)

The two balls are in contact for 0.50 s.
The average contact force acting on ball A is 4.0 N.
Air resistance is assumed to be negligible.

(a) State and explain the magnitude and direction of the average contact force acting on B.

……

……

…… [2]

(b) The total impulse acting on ball A during contact is not equal in magnitude to the total impulse acting on ball B.

(i) State whether the total momentum of the system of two balls is conserved.

…… [1]

(ii) Determine the magnitude of the velocity of A at the instant the two ball separates.

magnitude of velocity of A = …… m s^{-1} [4]

[Total: 7]

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3 Compton (1923) demonstrated that light behaves like a stream of discrete particles that carry both energy and momentum by scattering X-rays off a light element.

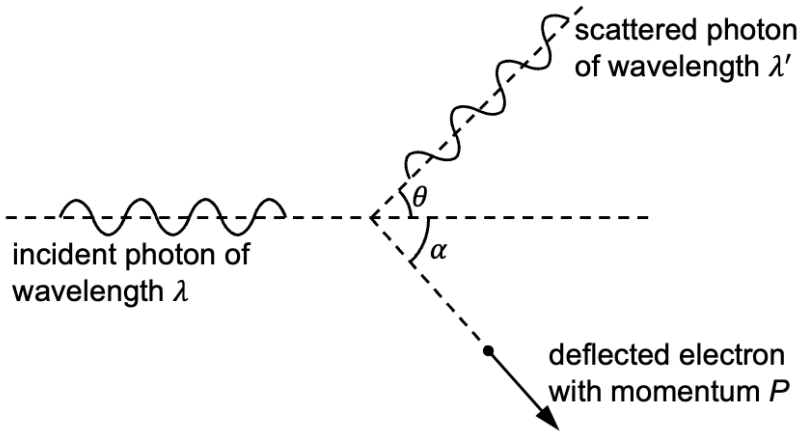


Fig. 3.1

By assuming energy and momentum are conserved in an elastic collision between an incident X-ray photon of wavelength λ and a stationary "free" electron (Fig. 3.1), he derived the equation for the scattered X-ray photon of wavelength λ':

$$\lambda' - \lambda = \frac{h}{mc}(1 - \cos\cos\ \theta\ )$$

where m is the rest mass of the electron, c is the speed of light in free space and θ is the angle of scattering of the photon.

In one experiment, the incident photon has a wavelength of 191.9×10^{-12} m and θ is 64.3^{o}.

(a) Show that the wavelength of the scattered photon is 193.3×10^{-12} m.

[1]

(b) By using conservation of energy, show that the momentum P of the deflected electron is 3.698×10^{-24} N s.

[3]

(c) Momentum is a vector quantity. By taking components of momentum at right-angles to the direction of the incident photon, calculate the angle α of deflection of the electron as shown in Fig. 3.1.

Explain your working.

α = ... ^{o} [4]

[Total: 8]

4 The planet Saturn has the most extensive and complex ring system of any planet in the Solar System. The rings have varying width and thickness. They are made up of mainly ice particles, with a trace of rocky material which are in orbit around Saturn. One of the rings, the D-ring, has an inner radius of
6.7×10^7 m.

(a) An object in krono-stationary orbit about Saturn has the same orbital period as the rotational period of Saturn. The rotational period of Saturn is 10 hours and 14 minutes.

A particle on the inner circumference of the D-ring has a speed of 2.38×10^4 m s^{-1}.

Deduce whether the particle is in krono-stationary orbit about Saturn.

……

…… [2]

(b) By considering the orbit of the particle in **(a)** show that the mass of Saturn is $5.7 \cdot 10^{26}$ kg.

[3]

(c) Saturn's largest moon, Titan, is the second largest moon in the solar system.

Titan is 1.2×10^6 km from Saturn. The mass of Titan is 1.4×10^{23} kg.

Determine the distance from the centre of Titan where the resultant gravitational field strength between Titan and Saturn is zero.

distance = …….. m [3]

[Total: 8]

5 A microwave oven, shown in Fig 5.1, consists of a reflective stainless steel casing, microwave source which is placed on one of the walls of the casing, and also a turntable at the bottom that rotates. The source produces microwave of frequency 2.45 GHz within the oven which is reflected within the casing forming a stationary wave.

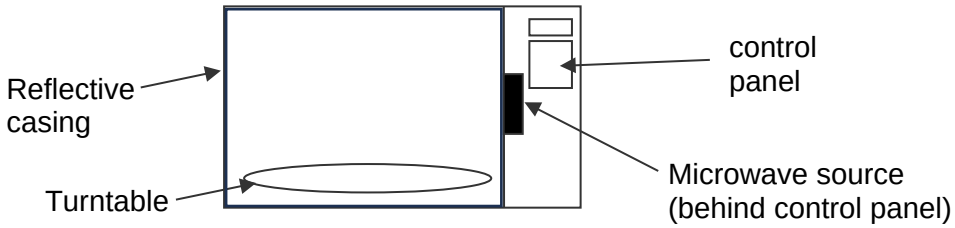


Fig. 5.1

(a) Describe three features of stationary wave.

1 .....

.....

2 .....

.....

3 .....

..... [3]

(b) Explain how stationary waves are formed inside the microwave oven.

.....

.....

.....

..... [2]

(c) A tray of mini marshmallow is arranged in a single layer, as shown in Fig. 5.2. The tray is placed in the microwave oven with the turntable removed. The oven is switched on to cook the marshmallow.



Fig. 5.2

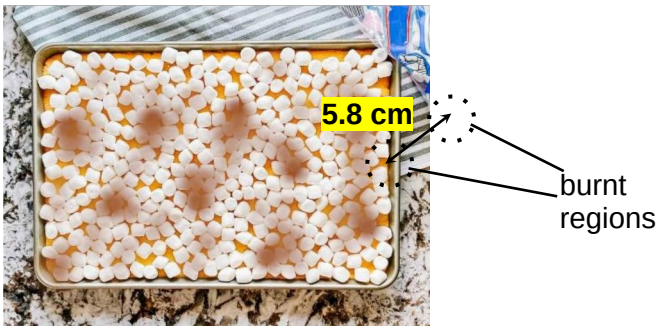


Fig. 5.3

Fig. 5.3 shows the tray of marshmallow taken out from the oven after cooking. It is observed that some regions are burnt more than the other regions. The average length between two adjacent burnt regions is 5.8 cm.

Determine the speed of microwave.

speed = …… m s^{-1} [2]

(d) The door of the microwave oven has a sheet of stainless steel with holes, with diameter about 2 mm, behind the glass, as shown in Fig. 5.4.



Fig. 5.4

Suggest a reason why microwave in the oven cannot pass through the glass while visible light in the oven can pass through.

……

…… [1]

[Total 8]

6 Fig. 6.1 shows a circuit containing a uniform wire P and a thermistor T that has a negative temperature coefficient.

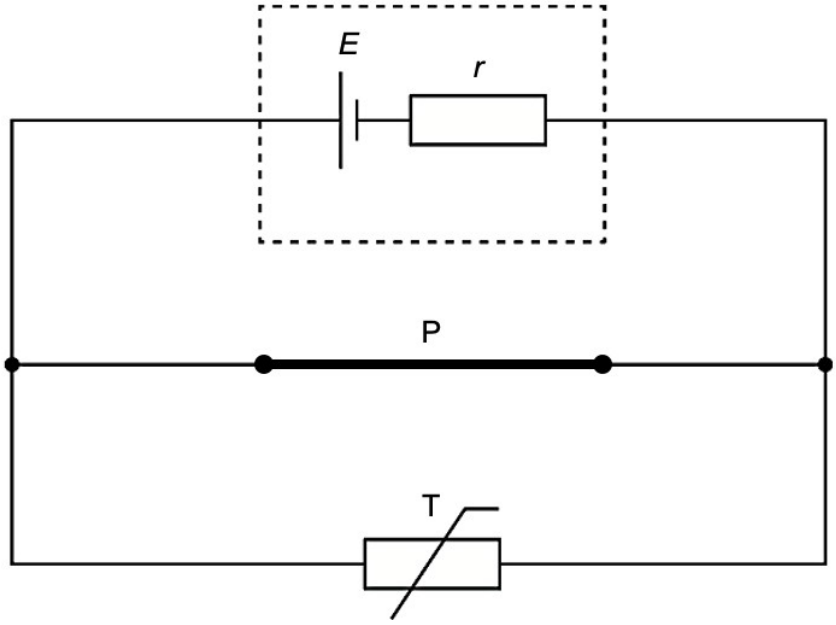


Fig. 6.1

(a) With reference to the current in the cell, explain why the current in **wire P** decreases with increasing temperature of the thermistor.

……

……

……

……

…… [3]

(b) The electromotive force (e.m.f.) *E* of the cell in Fig. 6.1 is 1.50 V.
The internal resistance *r* of the cell is 0.12 Ω.
The resistance of wire P is 6.00 Ω.
At a particular temperature of the thermistor, the current in P is 0.200 A.

Determine the resistance of the thermistor.

resistance = …….. Ω [3]

(c) Wire P has a length of 0.223 m.
The **total** number of charge carriers N in the wire is 1.10×10^{20}.

Calculate the average drift velocity of the charge carriers (electrons) in the wire.

average drift velocity = ……. m s^{-1} [2]

[Total: 8]

7 One possible nuclear reaction that takes place is

$$92235\,U + 01\,n \longrightarrow 4295\,Mo + 57139\,La + 2\ \ 01\,n + 7\ \ -10\,e$$

Some masses are shown in Fig. 7.1.

	mass / u
proton $(11\,p)$	1.007 276
neutron $(01\,n)$	1.008 665
uranium-235 nucleus $(92235\,U)$	235.043 930

Fig. 7.1

(a) **(i)** Show that the binding energy per nucleon of a uranium-235 nucleus is 7.409 MeV.

[3]

(ii) The nucleon number of an isotope of the element rutherfordium is 267.

State whether the binding energy per nucleon of this isotope will be greater than, equal to or less than the binding energy per nucleon of uranium-235.

…… [1]

(b) The binding energy per nucleon of molybdenum-95 $(4295\,Mo)$ and lanthanum-139 $(57139\,La)$ are 8.443 MeV and 8.189 MeV respectively.

Calculate the total energy, in MeV, released in this nuclear reaction.

energy = …….. MeV [2]

(c) The nuclei in 1.2×10^{-7} mol of uranium-235 all undergo this reaction in a time of 25 ms.

Calculate the average power release during the time of 25 ms.

power = …….. W [3]

[Total: 9]

8 Read the passage below and answer the questions that follow.

Breakthrough Starshot: Laser-Driven Interstellar Probes

Breakthrough Starshot is a proposed space exploration project that aims to send extremely small spacecraft, known as Starchips, to the Alpha Centauri star system. Unlike conventional spacecraft that carry large amounts of fuel, Starshot intends to use a powerful Earth-based array of lasers to accelerate a lightweight spacecraft attached to a reflective light sail, as shown in Fig. 8.1. The concept relies on radiation pressure exerted by photons from the laser beam. Although photons have no rest mass, they carry momentum, and when they are reflected from a surface, they transfer momentum to that surface, producing a force.



Fig. 8.1

The nearest star system, Alpha Centauri, is approximately 4.37 light-years from Earth. Conventional spacecraft would require tens of thousands of years to travel this distance. The Starshot proposal aims to accelerate a spacecraft to approximately 0.20c, where c is the speed of light. At this speed, the journey could potentially be completed in around 20 years.

A major component of the project is the laser array. The proposed array consists of numerous synchronized laser emitters operating together as a phased array. By controlling the phase of each emitter, the individual beams combine into a highly focused beam directed toward the sail, delivering a total power of 100 GW at the sail.

The radiation force on the sail depends on the laser intensity and the sail reflectivity. A higher power output increases the force and therefore the acceleration of the spacecraft. The array must maintain accurate beam collimation. As the spacecraft moves further from Earth, beam divergence can cause the laser spot to spread, reducing the intensity incident on the sail. A larger array aperture reduces diffraction and helps maintain beam focus over long distances.

Atmospheric effects also influence the design. Turbulence in the Earth's atmosphere can distort the laser beam, reducing efficiency. Special adaptive optics systems may be required to compensate for atmospheric distortion. In addition, the array must dissipate heat generated by the laser equipment and maintain precise alignment among thousands of individual laser elements. The energy consumption of

the laser facility is expected to be extremely high, requiring substantial power generation and storage infrastructure.
One of the critical components of the spacecraft is the light sail. The sail must be extremely thin and lightweight to maximise acceleration according to Newton's second law. At the same time, it must possess high reflectivity so that most of the incident laser energy is reflected rather than absorbed. Reflection provides momentum transfer while minimising heating.

Material selection for the sail is therefore crucial. A highly reflective dielectric material may have a reflectivity exceeding 99.9% at the operating laser wavelength of 1.0 μm. Nevertheless, a small fraction of the laser energy will always be absorbed. If the absorbed energy is not sufficiently removed, the sail temperature may rise to the point where structural failure occurs.

The rate of thermal energy absorbed P_{abs} by the sail can be calculated using

$$P_{\text{abs}} = (1 \quad R) P_{laser}$$

where *R* is the reflectivity and P_{laser} is the incident laser power.

To prevent overheating, the sail radiates thermal energy into space. Assuming the sail behaves approximately as a blackbody radiator with emissivity ε, the power radiated P_{rad} by the sail is described by the Stefan-Boltzmann law,

$$P_{rad} = \varepsilon \sigma A T^4$$

where

- $\sigma = 5.67 \times 10^{-8} W m^{-2} K^{-4}$ is the Stefan-Boltzmann constant,
- A is the total radiating area,
- T is the thermodynamic temperature of the sail,

If both sides of the sail radiate effectively, the total radiating area is twice the geometric area of the sail.

At thermal equilibrium,

$$P_{\text{abs}} = P_{rad}$$

This condition allows engineers to determine the operating temperature of the sail and assess whether the chosen material can survive the acceleration phase.

The sail must also remain mechanically stable. Uneven heating could produce thermal stresses that deform the sail and alter the direction of reflected laser light. Such deformation might reduce acceleration efficiency or even destabilise the spacecraft. Engineers must therefore optimise the sail thickness, shape, material properties and thermal conductivity.

The acceleration phase lasts only a few minutes. During this period, the spacecraft experiences extremely large accelerations. The onboard electronics must survive these forces while maintaining minimal mass. Since any increase in mass decreases acceleration for a given radiation force, all components must be carefully miniaturised.

A miniature power source is needed to power the camera, sensors and communication laser. No specific flight-ready power source has been identified. The mission concept assumes an ultra-lightweight power system of a few milligrams, likely involving advanced micro-batteries, radioisotope power, or another future technology that has not yet been fully developed.

[Turn over

(a) The laser beam hits the light sail at 90° and is totally reflected.

Starting from Newton's second law, show that the force exerted by the beam on the light sail is about 670 N.

[3]

(b) The spacecraft and sail have a total mass of 4.0 g.

(i) Determine the acceleration of the spacecraft.

acceleration = …… m s^{-2} [1]

(ii) Determine the time taken for the spacecraft to accelerate to 0.20*c* from an initial speed of zero.

time = …… s [1]

(c) The spacecraft would have travelled approximately 1.0×10^{10} m during the acceleration phase.

The sail is circular and has a diameter of 5.0 m.

(i) Show that the angular width of the sail from Earth at the end of the acceleration phase is 5.0×10^{-10} rad.

[1]

(ii) Explain why it is not desirable for the beam width of the laser to be significantly greater than 5.0×10^{-10} rad.

……..

…… [1]

(iii) If the laser beam comes from a single laser source, determine the diameter of the aperture of the laser source needed to restrict the beam width to 5.0×10^{-10} rad.

diameter = ... m [1]

(d) The light sail reflects 99.9% of the laser light incident on it. The emissivity ε is 0.80.

Determine the equilibrium temperature of the sail.

temperature = ... K [3]

(e) Suggest a reason why we do not consider heat loss through conduction and convection.

.....

..... [1]

(f) **(i)** The angular width of the sail from Earth is 1.0×10^{-6} rad at the start of the acceleration phase.

With reference to the answers to **(c)**, explain why the laser beam produces an uneven heating across the surface of the sail.

.....

..... [1]

(ii) Explain how the uneven heating across the surface of the sail can be reduced throughout the entire acceleration phase.

.....

.....

.....

..... [2]

(g) One of the options for an onboard power source is plutonium-238. Each plutonium-238 nucleus emits one α-particle of energy 5.48 MeV, of which only 6.7% can be harnessed as electrical energy.

The half-life of plutonium-238 is 86.4 years or 2.72×10^9 s.

The spacecraft requires at least 5.0 mW of power.

Determine the initial mass of plutonium-238 needed such that it can still supply this amount of power after 30 years.

mass = …… mg [5]

[Total: 20]

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